Up to which $k_{\max}$ is linear galaxy bias enough?

A linear-bias fit to a one-loop truth on four bin layouts

November 2026

1 Summary

The question. The forecast fits the DESI galaxy clustering band powers C_ℓ^{gg} of thin redshift bins with linear galaxy bias, $\delta_g = b_1\,\delta$: the galaxy density contrast δ_g is the matter density contrast δ times one number b_1 per bin. This is exact on large scales, but higher-order bias terms grow with the wavenumber k, so linear bias holds only below some scale cut $k_{\max}$. For the four linear-bias priors of the correlated-prior note, `notes/correlated_bias_prior_v2`, this note asks:

1. Up to which $k_{\max}$ does a linear-bias fit to a one-loop truth stay unbiased, on each of four bin layouts?

2. Does the accuracy test of the linear-bias priors give the same verdict for Ω_m and A_s as for (w_0, w_a)?

Here $w(a) = w_0 + w_a(1-a)$ is the dark-energy equation of state, with a the scale factor; Ω_m is the matter density today in units of the critical density; and A_s is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$.

The answer.

1. Linear bias is safe only at the smallest scanned cut, $k_{\max} = 0.05\,h/\mathrm{Mpc}$, and for every prior and every truth only on the layout of equal comoving width, where it keeps 1–6% of the dark-energy figure of merit. At $0.075\,h/\mathrm{Mpc}$, the cut of the correlated-prior note, a linear fit is biased on every layout under every prior, a free bias per bin included, so that note fits the one-loop model A there.

2. Yes. The test excludes the growth-tied prior, finds the baseline prior honest only under its own scatter, and the pessimistic prior honest under both.

The setup. The data are the DESI C_ℓ^{gg} alone, computed without the Limber approximation, for six tracer groups (BGS, LRG1–3, ELG1–2) on four bin layouts: equal comoving width, $\Delta d_{\mathrm{com}} = 62$ Mpc, and fixed redshift width, $\Delta z = 0.02$, 0.05 and 0.2. The first two are the thin layouts.

With $D(z)$ the linear growth factor ($D(0) = 1$), $\bar{z}_i$ the mean redshift of bin i and B_g one amplitude per group g, the four priors are: **growth-tied**, $b_i = B_g/D(\bar{z}_i)$; **baseline**, a free amplitude and tilt per group plus a residual of width $\sigma_g = 3\%$ for BGS and 1% elsewhere, correlated over $L = 0.2$ in redshift; **pessimistic**, $\sigma_g = 3\%$ everywhere and $L = 0.05$; and

free, one flat-prior bias per bin. The covariance is Gaussian, evaluated at the calibrated truth defined below.

A fit's shift is its Fisher linear response (Appendix A), in units of its marginalized error ellipse for (w_0, w_a) and of its marginalized error σ for Ω_m and A_s. A fit is unbiased when every shift is below $\sigma/3$, the correlated-prior note's threshold.

The scale cut. $k_{\max}$ labels one cut per bin: bin i is cut at $k_{\max}/D(\bar{z}_i)$, since matter is less nonlinear at higher redshift. The cut is the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [1], after DeRose and Chen [2]. Each band power splits into $C_\ell^{\rm IR}$, from the power spectrum below the bin's cut, and $C_\ell^{\rm UV}$, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; \alpha = 0.05 \tag{1}$$

and within the shipped mask, which caps every bin at $k = 0.1185\,\mathrm{Mpc}^{-1}$ ($0.176\,h/\mathrm{Mpc}$) with no growth scaling. The scan runs over $k_{\max} = 0.05, 0.075, \ldots, 0.2\,h/\mathrm{Mpc}$, each the $z = 0$ value.

The truths. The truth is the model A of the correlated-prior note, computed with CosmoLike's one-loop engine; that note also holds model B and the largest cut at which model A itself holds, $0.075\,h/\mathrm{Mpc}$ on the thin layouts. Its real-space galaxy power spectrum is

$$P_{gg}(k) \;=\; b_1^2\,P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_1, b_2, b_{s^2}) \;+\; \alpha_0\,k^2 P_{\rm mm}(k), \tag{2}$$

with $P_{\rm mm}$ the nonlinear matter power spectrum and $P_{\rm loop}$ the one-loop terms of the quadratic bias b_2 (the response of the galaxy density to δ^2) and the tidal bias b_{s^2} (its response to the squared tidal field). The k^2 counterterm α_0, in $h^{-2}\,\mathrm{Mpc}^2$, absorbs the small-scale sensitivity of the loops.

The *calibrated* truth keeps the fiducial b_1, puts b_2 and b_{s^2} on their calibrations (Appendix A.1), and sets α_0, the third-order bias b_3 and model A's stochastic terms to zero. Four brackets offset every group's b_2 by ± 1 ("$b_2 \pm 1$", model A's prior width) or set $\alpha_0 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$, the counterterm prior width of the DESI 2024 full-shape analysis [5]. "Every truth" means all five.

The safe cut, layout by layout. The safe $k_{\max}$ is the largest scanned cut at which every shift of every truth considered stays below $\sigma/3$, there and at every smaller cut with a fit. Table 1 lists it; Figure 1 draws the shifts.

- $\Delta d_{\rm com} = 62$ Mpc: the largest shift at 0.05 is 0.14σ.
- $\Delta z = 0.02$: the criterion keeps no band at 0.05, so the first fit is at 0.075, where every prior fails.
- $\Delta z = 0.05$: at 0.05 the largest shift is 0.28σ under the growth-tied and baseline priors. The pessimistic prior fails the brackets $b_2 - 1$ and $\alpha_0 = -12.5$ ((w_0, w_a) shifts 0.42 and 0.44), and the free prior fails already for the calibrated truth (0.36).
- $\Delta z = 0.2$: at 0.05 the same two brackets shift (w_0, w_a) by up to 0.50 (growth-tied) and 0.59. Its groups hold one or two bins, so the free amplitude and tilt already free every bin, and the baseline, pessimistic and free priors coincide.

At 0.05 the largest shift always comes from a negative bracket.

Layout	growth-tied	baseline	pessimistic	free
$\Delta d_{\rm com} = 62$ Mpc	0.05 / 0.05	0.05 / 0.05	0.05 / 0.05	0.05 / 0.05
$\Delta z = 0.02$	fails / fails	fails / fails	fails / fails	fails / fails
$\Delta z = 0.05$	0.05 / 0.05	0.05 / 0.05	0.05 / fails	fails / fails
$\Delta z = 0.2$	0.05 / fails	0.05 / fails	0.05 / fails	0.05 / fails

Table 1: The largest safe $k_{\rm max}$ [h/Mpc] of a linear-bias fit to the one-loop truth, as "calibrated truth / every truth"; "fails": the smallest fitted cut already fails. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

At $k_{\rm max} = 0.075\,h$/Mpc every prior is biased. For the calibrated truth the largest shift ranges from 0.39σ ($\Delta z = 0.2$, growth-tied) to 1.8σ ($\Delta d_{\rm com} = 62$ Mpc). On $\Delta d_{\rm com} = 62$ Mpc the four priors move Ω_m by 1.5–1.8σ, A_s by 0.95–1.2σ and (w_0, w_a) by 0.27–0.54, 6 to 52 times their shifts at 0.05.

Why more bias freedom does not help. A linear bias multiplies a bin's power spectrum by the constant b_i^2 at every k, while the loop terms and the counterterm of Eq. (2) change shape with k. No choice of b_i removes that scale dependence, so the fit pushes it into the cosmological parameters, which do change the shape of the band powers; freeing the biases changes which parameter takes it, not whether one does. On $\Delta d_{\rm com} = 62$ Mpc at 0.075, going from the growth-tied to the free prior lowers the Ω_m shift from 1.77 to 1.50 and raises the (w_0, w_a) shift from 0.27 to 0.54. On $\Delta z = 0.05$ the baseline prior moves the error from Ω_m (0.19, against 1.28 for growth-tied) to (w_0, w_a) (0.98, against 0.29).

The accuracy test for Ω_m and A_s. Figure 2 repeats the correlated-prior note's accuracy test with linear-bias fits and linear truths at $k_{\rm max} = 0.075\,h$/Mpc. Each truth is a known bias drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population elsewhere) plus a residual drawn from the baseline or the pessimistic prior: 2000 draws, each fitted under every prior. A fit is scored by $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, with χ^2 the squared distance of the recovered parameters from the truth in units of the fit's marginalized covariance, $n_{\rm dof} = 2$ for (w_0, w_a) and 1 for Ω_m or A_s, and $\langle\cdot\rangle$ the average over the random truths and the data noise. η measures how well each fit's quoted error bar covers where the fit actually lands: 1 for an honest error bar and, above 1, the factor by which it is too small. It scores the parameters, not the χ^2 of the data: a biased fit can match the band powers while its cosmology is wrong.

The verdict for Ω_m and A_s is that for (w_0, w_a):

- **growth-tied** is excluded: the conserved-population drift alone shifts (w_0, w_a) by 5.1–11.6 and Ω_m by 0.55–6.1σ across the layouts;
- **baseline** gives $\eta \leq 1.02$ under its own scatter, but under pessimistic scatter its error bars are too small by up to 1.65 for (w_0, w_a), 2.0 for Ω_m and 1.48 for A_s ($\Delta z = 0.05$);
- **pessimistic** is honest or conservative under both, with $\eta \leq 1.05$;
- **free** recovers every such truth exactly.

Every value is in `plots/ladder_bias_params/data.json`.

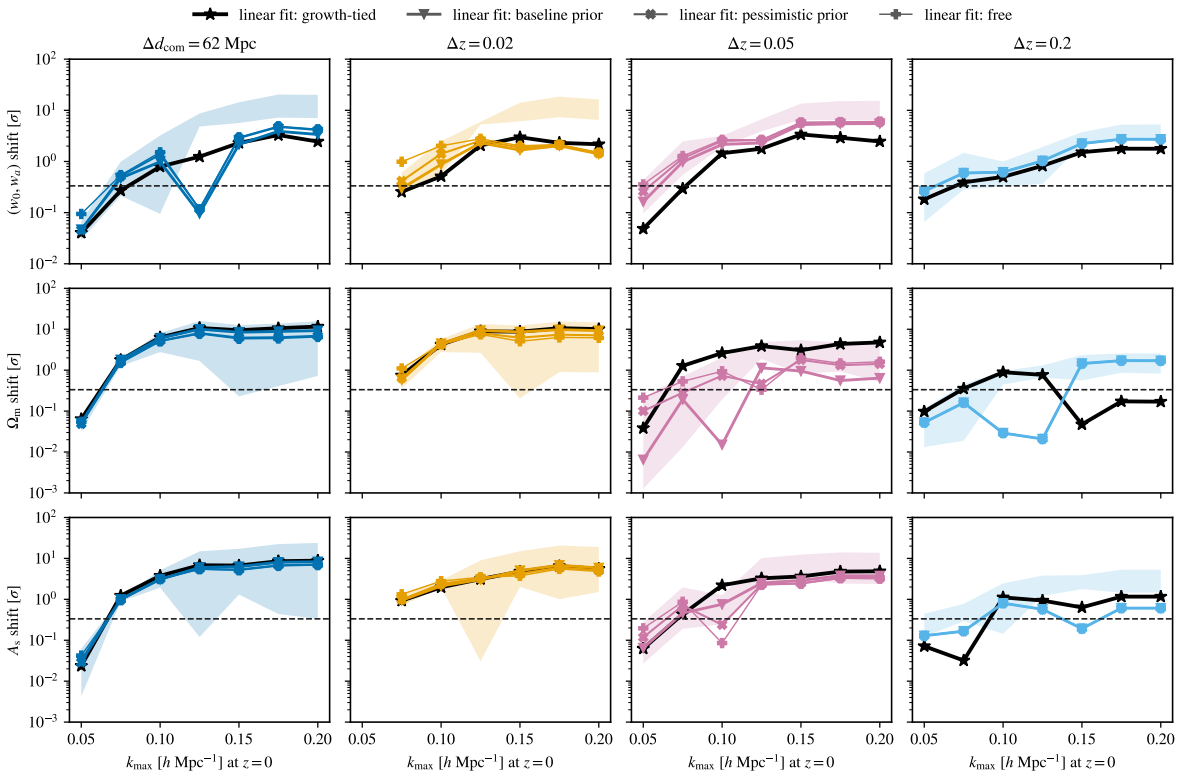


Figure 1: The shift of a linear-bias fit to the one-loop truth against the scale cut, one column per layout. Lines: the calibrated truth; color: the layout (black: growth-tied); line width and marker: the prior (growth-tied thickest with stars, baseline down-triangles, pessimistic crosses, free thinnest with plus signs); band: the range of the four brackets, for the pessimistic prior only; dashed: $\sigma/3$. On $\Delta z = 0.02$ the cut at 0.05 keeps no band; on $\Delta z = 0.2$ the free line covers the baseline and pessimistic ones. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

The price of the safe cut. Figure 3 draws, for the three priors that pass the accuracy test, the dark-energy figure of merit ${\rm FoM}(w_0, w_a) = [\det {\rm Cov}(w_0, w_a)]^{-1/2}$, the inverse area of the marginalized (w_0, w_a) ellipse, against $k_{\rm max}$. On $\Delta d_{\rm com} = 62$ Mpc the cut at 0.05 keeps 5% of the FoM at 0.075 under the baseline prior (0.26 against 5.1), 6% under the pessimistic prior and 1% under free. The loss is part of why the cut passes: it keeps fewer bands, on larger scales, where the one-loop terms are smaller and the data less precise.

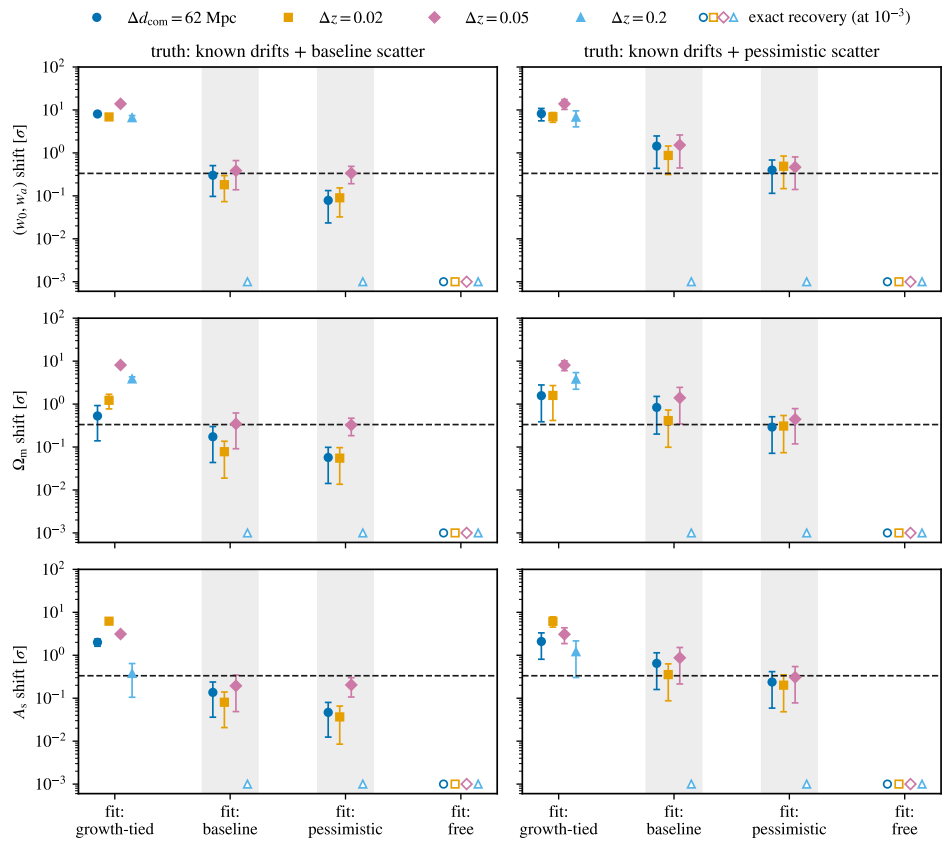


Figure 2: The accuracy test of the linear-bias priors, scored for (w_0, w_a), Ω_m and A_s: 2000 truths, the known drifts plus a residual drawn from the baseline (left) or the pessimistic prior (right), each fitted under every prior. Points: the mean shift over the truths; bars: 16th–84th percentiles; color and marker: the layouts; gray columns: the baseline and pessimistic priors; dashed: $\sigma/3$. Where the fit's prior equals the truth's scatter, the fit's error covers the mean shift and $\eta = 1$. $k_{\rm max} = 0.075\,h/{\rm Mpc}$; $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

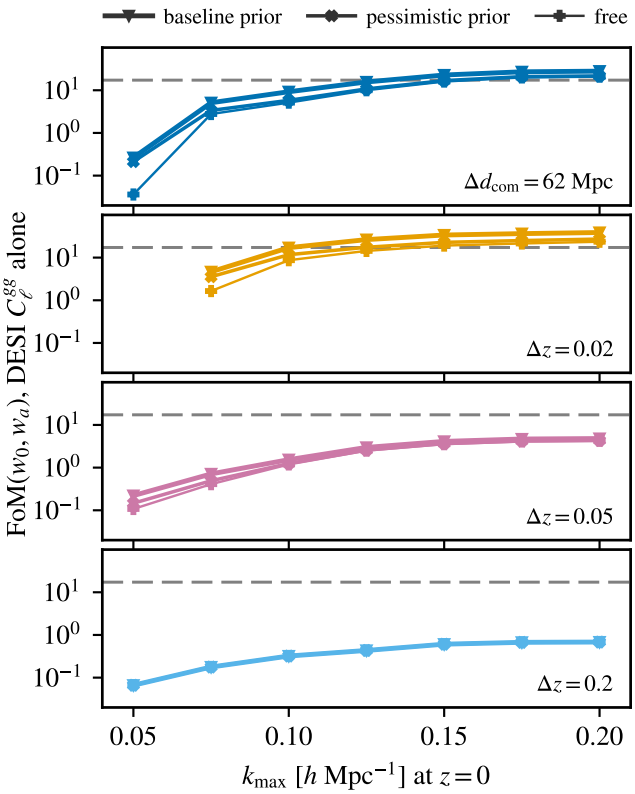


Figure 3: FoM(w_0, w_a) of a linear-bias fit to the DESI C_ℓ^{gg} alone against the scale cut, one panel per layout, for the priors that pass the accuracy test. Color: the layout; line width and marker: the prior; gray long-dashed: LSST Y1 3x2pt. On $\Delta z = 0.2$ the three priors coincide. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

A Method

A.1 The truths

Each truth is a full model A data vector at the fiducial cosmology, with $b_{1,i} = B_g/D(\bar{z}_i)$. The calibrations are the Lazeyras et al. separate-universe fit [3], $b_2(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$, and $b_{s^2} = -\frac{4}{7}(b_1 - 1)$, the co-evolution of a population formed with local bias and moved by gravity [4]. The brackets $b_2 \pm 1$ set every group's offset $\Delta_{2,g} = b_2 - b_2(b_1)$ to ± 1; $\alpha_0 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$ gives $\alpha_0 k^2 = 0.5$ at $k = 0.2\,h/\mathrm{Mpc}$, a 50% correction there [6]. CosmoLike's counterterm slot is $2\,b_1 b_K\,k^2 P_{\mathrm{mm}}$, with b_K its per-bin derivative-bias coefficient, so $\alpha_0 = 2\,b_1 b_K\,(c/H_0)^2$; DESI multiplies the linear spectrum instead.

The vectors use the accuracy settings of the cached linear derivatives. The low-rank non-Limber engine factorizes $P_{gg} - P_\infty$, with P_∞ the negative white-noise constant that the subtracted loop terms tend to at high k; P_∞ reaches C_ℓ through Limber, exact for a constant (`notes/higher_order_bias`, subsection "The white-noise limit of the renormalized loops").

A.2 The fit: linear response under each prior

Let m be the expected data vector, C its covariance on the kept entries and $\mathsf{D} = \partial m/\partial p$ the derivatives of the linear-bias model with respect to the fitted parameters: the seven cosmological parameters ($10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0, w_a; flat priors), the bias parameters of the prior and the per-group velocity dispersions (Gaussian priors). The Fisher matrix is $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D} + F_{\mathrm{priors}}$, with F_{priors} the velocity-dispersion priors plus, for the baseline and pessimistic priors, the projected precision of the bias prior (the correlated-prior note, sections "The projected precision for two bins" and "The memory model").

C is the Gaussian covariance of the calibrated truth, one-loop terms included, with massless neutrinos (its signal sits 3% above the data vector's), computed for 14,000 square degrees with its inverse rescaled to the survey's 12,300.

If the data are m_{truth} while the model at the fiducial predicts m_{lin}, to first order the best fit moves by

$$\delta p \;=\; F^{-1}\,\mathsf{D}^\mathsf{T}\,\mathsf{C}^{-1}\big(m_{\mathrm{truth}} - m_{\mathrm{lin}}\big)\,: \tag{3}$$

the data pull on the old best fit with the gradient $\mathsf{D}^\mathsf{T}\mathsf{C}^{-1}(m_{\mathrm{truth}} - m_{\mathrm{lin}})$, and the curvature F turns that pull into a displacement. The growth-tied prior uses the cached growth-tied derivatives, one bias column per group, and the others the per-bin derivatives, both from the linear notebook, `EXAMPLE_EVALUATE_FISHER.ipynb`.

A.3 The scores and the threshold

With δw the (w_0, w_a) entries of δp and C_w the (w_0, w_a) block of F^{-1}, the (w_0, w_a) shift is

$$\Delta_w \;=\; \sqrt{\delta w^\mathsf{T}\,\mathsf{C}_w^{-1}\,\delta w}, \tag{4}$$

the displacement of the dark-energy ellipse in units of its own size. For one parameter p the shift is $\Delta_p = |\delta p|/\sigma_p$, with $\sigma_p = \sqrt{(F^{-1})_{pp}}$. A shift of $\sigma/3$ added in quadrature to a 1σ error gives $\sqrt{1 + 1/9} = 1.05$: below the threshold the total error grows by at most about 5%.

A.4 Cuts without data

At each $k_{\max}$ the inverse covariance is rebuilt on the kept entries; at the largest cuts the shipped mask sets more and more of them, and the curves of Figures 1 and 3 flatten. Bins that lose every band keep only finite-difference noise in their bias columns, so a parameter whose information (data and prior) at a cut is below 10^{-10} of that on the full shipped mask leaves the fit. A cut that leaves no data for the cosmological parameters, as on $\Delta z = 0.02$ at $0.05\,h/\mathrm{Mpc}$, gives no fit, and the search for the safe $k_{\max}$ skips it.

B Numerical checks

- **One fiducial vector.** The linear vector computed with the truths equals the cached reference of the linear notebook exactly on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$ and 0.05. On $\Delta z = 0.2$ the largest relative difference, 4.5×10^{-3}, sits at a cross spectrum with nearly zero band powers; the median is 2×10^{-5}, and $\Delta\chi^2 = 2 \times 10^{-5}$ on the shipped mask.

- **Size of the one-loop sector.** The one-loop part of the calibrated truth is a median 1.9%, 1.6%, 2.3% and 1.9% of the linear band powers on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2.

- **The truths of the accuracy test.** They use the linear notebook's seed and order, so they are the truths the correlated-prior note fits under model A. When the fit's prior equals the truth's scatter, the scatter and noise terms of $\langle\chi^2_w\rangle$ sum to exactly 2 (here to 6×10^{-11}), and η of (w_0, w_a) is within 0.03 of 1, the remainder being the drift.

- **The covariance grid.** The covariance's non-Limber grid (`nonlimber_accuracyboost` 2) is converged to within 2% on the adjacent thin-bin cross spectra and 0.4% on the diagonal.

- **The white-noise constant.** Its exact projection leaves linear-bias vectors bitwise unchanged (`notes/higher_order_bias`).

C Limitations

- **Linear response.** The shifts are the first-order response of Eq. (3), not a refit. Most verdicts rest on shifts well below the threshold or several times above it; a refit could move those near it, on $\Delta z = 0.05$ and 0.2 at 0.05 and growth-tied on $\Delta z = 0.2$ at 0.075 (0.39σ).

- **Theory truths, not mocks.** Higher-order biases outside the brackets are not covered.

- **Negative power in one bracket.** For BGS-like b_1 the $\alpha_0 = -12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth makes P_{gg} negative above $k \approx 0.37\,h/\mathrm{Mpc}$, where the low-rank engine floors $P_{gg} - P_\infty$; this is immaterial at $k_{\max} \leq 0.075\,h/\mathrm{Mpc}$.

D Open issues

- **The covariance's one-loop terms enter through Limber.** Its low-ℓ non-Limber recompute is linear, an error of at most 1.3% of C_ℓ in the lowest bands of thin bins.
- **The cut was built on the linear model.** The counterterm, with its k^2 weight, draws more of its signal from beyond the cut: on $\Delta z = 0.05$ at $0.075\,h/\mathrm{Mpc}$ the ultraviolet fraction of the α_0 template is a median 0.076 on the kept bands, above α, against a median 0.017 and at most 0.052 for the full model A vector (the correlated-prior note, "Open issues").
- **No stochastic term in the truths.** The cross spectra of adjacent thin bins are sensitive to small-scale radial physics at their shared edge, which model A's term $\mathrm{SN}_2\,k^2\mu^2$ carries (μ the cosine to the line of sight). Every truth here sets it to zero.
- **No redshift-space loops.** Truths and fits carry linear Kaiser distortions and the Fingers-of-God smearing of each bin's redshift distribution, but no redshift-space one-loop terms, which CosmoLike lacks.
- $b_3 = 0$**.** CosmoLike's third-order bias template is cosmology-independent and wrong.

References

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[3] T. Lazeyras, C. Wagner, T. Baldauf and F. Schmidt, "Precision measurement of the local bias of dark matter halos", 2016, arXiv:1511.01096.

[4] S. Saito, T. Baldauf, Z. Vlah, U. Seljak, T. Okumura and P. McDonald, "Understanding higher-order nonlocal halo bias at large scales by combining the power spectrum with the bispectrum", 2014, arXiv:1405.1447.

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